Jun-Yu Pan

139-0663-0737 | panjunyu@sjtu.edu.cn | albertmentat.github.io | github.com/AlbertMentat

Education

Shanghai Jiao Tong University September 2026 – March 2029 (Expected)
M.Eng. in Electronic Information, SJTU Paris Elite Institute of Technology *Shanghai, China*

- Advisor: Prof. Wei-Long Zheng
- Research interests: EEG foundation models, multimodal large language models (MLLMs), and applications of EEG foundation models

Shanghai Jiao Tong University September 2022 – June 2026
Bachelor of Engineering, SJTU Paris Elite Institute of Technology *Shanghai, China*

- Weighted Average: 87.12/100 | GPA: 3.66/4.3 | Rank: 7/19
- Honors: Outstanding Bachelor's Thesis (Top 1%); Third-Class Scholarship (Top 15%); Academic Progress Scholarship
- Second Prize, "Shan Shu" Mathematical Contest in Modeling; Shanghai Third Prize, Contemporary Undergraduate Mathematical Contest in Modeling

Research Experience

Microsoft Research Asia (MSRA) April 2025 – Present
Research Intern | Advisor: Yansen Wang *Shanghai, China*

- Research on data-centric EEG foundation models and EEG-MLLM alignment, spanning pretraining-data curation, multi-dataset fine-tuning, and generative visual grounding.
- Completed three manuscripts on EEG foundation models and MLLMs, including one first-author arXiv paper and two NeurIPS 2026 submissions.

BCMI, Shanghai Jiao Tong University June 2023 – June 2026
Research Assistant | Advisor: Prof. Wei-Long Zheng *Shanghai, China*

- Studied EEG-based affective computing, with an emphasis on cross-scenario, contrastive, and multimodal emotion recognition.
- Developed DDCT for video-to-game emotion recognition and M4Lego for multimodal gameplay emotion recognition using EEG and eye movements.

Selected Research Projects

Generative Visual Grounding for Universal EEG Understanding *Microsoft Research Asia*

- Proposed Generative Visual Grounding (GVG), which uses an EEG-to-image generator to create instance-specific proxy images and maps EEG into discrete visual tokens so that MLLMs can exploit pretrained visual priors.
- Built GVG-X-Omni and GVG-Janus with image-only and Image+Text alignment; matched a 1.7B-parameter text-aligned baseline while tuning 170M parameters on a frozen 7B backbone, and achieved further gains through trimodal alignment.

Controlled Audit of Data Scaling in EEG Foundation Models *Microsoft Research Asia*

- Ran a controlled audit of 151 pretraining configurations across LaBraM, CBraMod, and BIOT, covering 16 public datasets and 13 downstream tasks.
- Developed codebook-coverage H-ordering and Pairwise Dataset Valuation (PDV) for data selection; informed subsets outperformed full-pool pretraining by up to 5.53 percentage points in balanced accuracy.

Selected Research Projects (continued)

Probe-Guided Multi-Dataset Fine-Tuning for EEG Foundation Models *Microsoft Research Asia*

- Developed POSE, a target-conditioned framework that selects and weights auxiliary datasets from MMD and Frechet-distance trajectories in a target-adapted feature space.
- Combined compatibility-aware selection with a warm-up, joint-training, and cool-down schedule; evaluation on 12 datasets and two EEG foundation models improved average target-only fine-tuning by 3–4% while mitigating negative transfer.

Cross-Scenario and Multimodal Emotion Recognition *BCMI, SJTU*

- Proposed Double Domain Converter Transformer (DDCT) for EEG emotion recognition across video and gameplay scenarios; the work was published at ICASSP 2025.
- Developed M4Lego, a multimodal pretraining framework that integrates EEG and eye movements using multi-level masking and Lego-style blocks for emotion recognition during gameplay.

Publications

[1] **Jun-Yu Pan**, Yansen Wang, Enze Zhang, Bao-Liang Lu, Wei-Long Zheng, and Dongsheng Li. "Visualizing the Invisible: Generative Visual Grounding Empowers Universal EEG Understanding in MLLMs." *International Conference on Learning Representations (ICLR)*, under review. arXiv:2605.18172.

[2] **Jun-Yu Pan**, Ruicheng Yin, Yansen Wang, Bao-Liang Lu, Wei-Long Zheng, and Dongsheng Li. "Which Data, Not How Much: A Controlled Audit of Data Scaling in EEG Foundation Models." *Advances in Neural Information Processing Systems (NeurIPS 2026)*, under review.

[3] Ruicheng Yin, **Jun-Yu Pan**, Yansen Wang, Xiaoqing Zheng, and Dongsheng Li. "Discern, Then Combine: Probe-Guided Multi-Dataset Fine-Tuning for EEG Foundation Models." *Advances in Neural Information Processing Systems (NeurIPS 2026)*, under review.

[4] **Jun-Yu Pan***, Jing-Yi Liu*, Bao-Liang Lu, and Wei-Long Zheng. "M4Lego: A Multi-Modal Pretraining Framework with Multi-Level Masking and Lego Blocks for Emotion Recognition During Gameplay." *IEEE Transactions on Affective Computing*, under review.

[5] **Jun-Yu Pan**, Hao-Long Yin, and Wei-Long Zheng. "Double Domain Converter Transformer for Improving EEG-Based Emotion Recognition from Video to Game Scenarios." *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2025.

*Equal contribution.

Technical Skills

Programming: Python, C **Frameworks:** PyTorch **Languages:** Mandarin, English, French